

QUESTION PAPER

Name of the Examination: WIN Semester/2023-24 – CAT1

Course Code: CSE3003

Course Title: COMPUTER NETWORKS

Set number: 3

Date of Exam: 06/02/2024 (FN)(B)

Duration: 90 mins

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. A student attaches a laptop to the campus Wi-Fi network and requests/receives a web page from www.vitap.ac.in. Explain the sequence of operations carried out with the help of different protocols used in the application, transport, network and link layers.

(10M)

Q2. i) A LAN system has 8 Systems using a Main switch which connects two sub-switches. Each Sub Switch Connects 2 hubs and each hub connects two systems. Show the complete connections.

(3M)

ii) Explain the reasons for which a switch cannot connect computers in different LAN setups. Switches store physical addresses in MAC address table. How physical addresses are utilized in data communication?

(7M)

Q3. i) For each of the following four network topologies, discuss the consequences if a connection fails.

(4M)

- a. six devices arranged in a bus topology
- b. four devices arranged in a dual ring topology
- c. five devices arranged in a mesh topology
- d. seven devices arranged in a star topology (not counting the hub)

ii) Explain three types of transmission impairment and justify with suitable scenarios.

(6M)

Q4. i) Explain the effectiveness of shield in shielded twisted pair. Explain the issues in twisted pair that are not present in coaxial cable.

(6M)

ii) State the factors of optical fiber helpful in reducing the effects of electromagnetic interference that is present in coaxial cable and twisted pairs.

(4M)

Q5. A digital bit stream with the following bits "1010101111001101" is to be broadcast.

Draw the waveforms shown below for the data that was transmitted.

(10M)

- a. Manchester
- b. Bipolar RZ
- c. Polar-NRZ-L
- d. Differential Manchester

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1, 2	1	1	10
Q2	1	1	1, 2	1	1	10
Q3	2	4	4	2	2	10
Q4	2	2	3,4	1	2	10
Q5	2	4	4	2	2	10



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QUESTION PAPER

Name of the Examination: WINTER 2023-2024 – CAT-1

Course Code: CSE3003

Course Title: Computer Networks

Set number: 7

Date of Exam: 07/02/2024 (FR) (C)

Duration: 90 Mins

Total Marks: 50 Marks

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

- Q1. A factory is developing a wireless control system for real-time monitoring and adjustment of robotic production lines. How can you implement a seven-layered data communication approach during data transmission between the production line and the control center? Explain the implementation at each layer in detail. **(10 Marks)**
- Q2. Picture yourself as a network engineer for a rapidly expanding startup company. The company needs a reliable and scalable network to support its expansion into a new office. As the network engineer, you have to design and implement a network solution for the new office, including choosing the right LAN, MAN, and WAN technologies. Discuss your network solution in detail with pros and cons. **(10 Marks)**
- Q3. A technology company is planning to upgrade its data center's network infrastructure. The data center needs a reliable, high-speed communication network to support a variety of services. Considering data security, higher speed with Higher Bandwidth and Lower Latency, which type of transmission media would be the optimal solution for this scenario? Explain your solution in detail. **(10 Marks)**
- Q4. A major international news organization hosts a live broadcast covering a historical peace summit. The organization is hosting the event from a new building and needs to deploy their own computer network. Discuss all suitable network topologies for each floor and the overall organization, considering their specific needs and challenges. Explain your choices with reasons. **(10 Marks)**

- Q5. Digital Communications Inc. is developing a new data transmission system. Given this scenario, evaluate the suitability of different encoding mechanisms for Digital Communications Inc.'s data transmission system. Draw the signal graph for the data stream "10110101101" using
- Non-Return-to-Zero-Level (2.5 Marks)
 - Polar Return to Zero (2.5 Marks)
 - Bipolar (2.5 Marks)
 - Manchester (2.5 Marks)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1, 2, 4	1 to 8	2,4	-	10
Q2	1	1, 2, 4	1 to 8	2,4	-	10
Q3	2	1, 2	1 to 7	2	-	10
Q4	2	1, 2	1 to 7	2	-	10
Q5	2	1, 2	1 to 7	2	-	10



QUESTION PAPER

Name of the Examination: Win 2023-24 Seniors Semester – CAT1

Course Code: CSE3003

Course Title: Computer Networks

Set number: 8

Date of Exam: 07/02/2024 (AN)

Duration: 90 min

Total Marks: 50

(C2)

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

- Q1.** A) A file transfer between two systems has resulted in corrupted data at the receiving end. Which layer of the OSI model is responsible for data formatting and encryption/decryption? Explain about the different functions of that layer in the OSI model. **(5M)**
- B) Starting from application layer till physical layer, write about how data is transformed and what header information is added in each layer of the OSI model? **(5M)**
- Q2.** A) Bridges and switches are layer two devices that use physical addresses. Explain the process of filtering in them. What is the characteristic difference between a bridge forwarding table and MAC address table? **(5M)**
- B) Hubs are multiport repeaters that are used in networks. However, it does not allow full duplex communications between multiple clients. Justify with suitable reasons. **(5M)**
- Q3.** Define the physical structure of twisted pair, coaxial and fiber optic cable and discuss about advantages, disadvantages, and performance in terms of attenuation/loss measured in dB/km. Explain which transmission media needs more repeaters out of twisted pair, coaxial and fiber optic cable. **(10M)**
- Q4.** A medium-sized company is planning to expand its network infrastructure to accommodate the growing number of employees and the increasing demand for seamless communication. The company has multiple departments spread across different floors of a building, and they need to ensure reliable connectivity, scalability, and ease of management.
- Question: Outline the considerations and factors you would take into account when deciding on the most suitable network topology for this company. Discuss the advantages and disadvantages of at four widely used network topologies that could meet the organization's requirements. Additionally, explain how your chosen topology enhances network performance, fault tolerance, and the ability to accommodate future expansion. **(10M)**

Q5. Consider the sequence of digital data 110110110011, and convert it into digital signal using following schemes: **(10M)**

- Unipolar RZ
- NRZ-I
- Manchester
- Bipolar AMI

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	3	1	2	1	10
Q2	1	4	1	2	1	10
Q3	2	2,4	1	2	1	10
Q4	2	2,3,4	4	-	1	10
Q5	2	2,3,4	4	-	1	10